

Data Processing: Proposal 1

Living Well? - A Global Data Analysis of Heat and Domestic Violence

[LivWell](#) is a dataset based on DHS surveys, including 1832 observations from 52 countries.

It includes:

- 14 indicators on women's wealth, status, etc.
- 20 indicators on domestic violence and women's decision-making power
- 129 climate indicators

Research Question:

"Is extreme heat related to increased violence against women?"

- Non-causal
- Primary question, next to exploratory ideas

Strategy:

- **Step 1:** Operationalise "Extreme heat," "violence against women," and "Women's Empowerment" as a moderator.
 - Load the csv, select the right rows and columns, etc.
 - Extreme heat and violence against women could be single variables.
 - For Empowerment, I want to create sub-indices from several variables and combine them into a total score.
- **Step 2:** Then, I want to analyse this using a fixed effects model and self-designed (but closely resembling bernoulli / tau tests) binary test as a robustness check.

- **Step 3:** Visualise the results and document everything on a website, ideally with interactive geo-plots.
- **(Optional Step 4):** *If* I have enough time, I would also like to illustrate potential intervention effects by operationalising mechanisms from the literature with LivWell variables. This would only be a *highly* approximate, exploratory “policy simulator”.

Note: ask if this works

Data sources are: LivWell. I will need *some* cleaning (more like selection). For Empowerment feature-creation, maybe PCA or a self-made scoring method. I will also need to deeply understand the data, visualise it, check whether it can be used for my analysis, etc. I want to visualise the results using GeoPandas.

Novelty: I will need to figure out how to do fixed effect regression in Python, how to use GeoPandas, potentially how to make visualisations on a website interactive. I will need at least two new libraries: GeoPandas and whatever I will use for the regression.